Fiscal Year:	FY 2009	**Task Last Updated:**	FY 01/25/2010
PI Name:	Law, Chung K. Ph.D.		
Project Title:	Structure and Response of Spherical Diffusion Flames		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Gaseous diffusion flames		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	08544	**Congressional District:**	12
Comments:	Chung Law's email is cklaw@princeton.edu		
Project Type:	Flight	**Solicitation / Funding Source:**	97-HEDS-01
Start Date:	10/01/2007	**End Date:**	09/30/2009
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	2	**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA GRC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Tse, Stephen (Rutgers University) Sacksteder , Kurt (NASA Glenn Research Center)		
Grant/Contract No.:	NNC06AA27A		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	The effort was divided into experimental and theoretical elements. The experimental effort consisted of using a porous spherical burner to issue fuel mixtures into an oxidizing environment. Different mixtures of fuel species permitted investigation of the phenomena of interest. The experiment effort was conducted primarily in the drop towers at the GRC and the NASA reduced-gravity aircraft. Space flight experiment was fully defined and approved during the program period. The theoretical effort consisted of the development of computational codes simulating the combustion phenomena of interest, and interpreting the calculated results on the basis of fundamental flame physics. The purpose of this flight definition task is to gain fundamental understanding of the structure and dynamics of diffusion flames in a simple, well-defined flow field. Using a spherical geometry this work considers: 1) gas-phase transient processes, including fuel vapor accumulation inside the flame and far-field diffusion, and their influence on flame movement and the achievement of steady flame behavior; 2) extinction boundaries of steady or near-steady flames and the		

	identification of the associated chemistry at the global, reduced, and detailed levels; 3) the characteristics of partially-premixed flames; 4) flamefront pulsation and cellular instability especially near the extinction limits; 5) the dynamics of rotating flames; and 6) the behavior and structure of quasi-spherical flames in partial gravity environments. These studies are intended to significantly enhance the understanding of diffusion flames, especially providing capability for predicting flame behavior under varied conditions and configurations. As such, this work should contribute to engineering approaches to practical combustion systems, achieving increased energy conversion efficiencies and reductions in unwanted combustion emissions in both terrestrial and extraterrestrial environments.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Enhanced understanding of flame-flow vortical interaction could lead to improved burner design and operation in terms of energy efficiency and pollutant emissions.
Task Progress:	The Science Concepts Review (SCR) for the project was passed in Feb 2008. Continual updates to the ACME Science Requirements Documents (SRD) have been made based on ongoing research. On-going ground-based investigation involve transient behavior of burner-supported spherical diffusion flames, structure and extinction characteristics of counterflow diffusion flames with flame radiation and non-unity Lewis numbers of the fuel and oxidant, the onset of near-limit flame oscillation with radiative heat loss in 1-D chambered planar flames using multi-scale activation-energy asymptotics, microgravity experiments on the effects of rotation on spherical diffusion flames with emphasis on varying Lewis numbers on both the fuel and oxidizer sides, and examination of the flame structure of low-buoyancy inverted spherical diffusion flames using laser-based diagnostics.
Bibliography Type:	Description: (Last Updated: 11/26/2025)
Articles in Peer-reviewed Journals	Yang F, Law CK, Sugn CJ, Zhang HQ. "A mechanistic study of Soret diffusion in hydrogen-air flames." Combustion and Flame 2010 Jan;157(1):192-200. http://dx.doi.org/10.1016/j.combustflame.2009.09.018 , Jan-2010